Empirical Analysis of Regulation and Cross-Currency Dynamics in Cryptocurrency Markets

1 Data

Our primary data source is depth-20 Level-2 order book snapshots and trade data from CoinAPI for the following exchanges: BinanceUS, Kraken, Coinbase, and Bybit. Not all exchanges offer all quote currencies, so coverage varies by venue. We also obtain USDC/USD and USDT/USD reference rates from CoinMetrics at 1-minute, hourly, and daily frequencies to normalize relevant metrics into USD-common units.

2 Comparison of BTC/USDT, BTC/USDC with BTC/USD

We measure cross-currency deviations between BTC/USD and BTC quoted in stablecoins ($c \in \{\mathrm{USDT}, \mathrm{USDC}\}$) using both top-of-book and full-depth L2 information. With ordered ask prices a and sizes v at level i $\{(a_{c,t,i}, v^a_{c,t,i})\}$ and bid levels $\{(b_{c,t,i}, v^b_{c,t,i})\}$, we define the mid-price $P^{\mathrm{mid}}_{c,t} = (a_{c,t,1} + b_{c,t,1})/2$ and the micro-price

$$P^{\mathrm{micro}}_{c,t} = \frac{a_{c,t,1} v^b_{c,t,1} + b_{c,t,1} v^a_{c,t,1}}{v^a_{c,t,1} + v^b_{c,t,1}},$$

which incorporates top-of-book depth imbalance. To capture executable prices, we compute depth-weighted execution prices by walking the book for trade size Q and define the depth-weighted mid-price as their average $P^{DW}_{c,t}(Q)$.

For $Q \in \{1, 5, 10\}$, we construct two complementary depth-weighted basis measures:

$$\mathrm{Basis}^{DW,\%}_{c,t}(Q) = \frac{P^{DW}_{c,t}(Q)}{P^{DW}_{\mathrm{USD},t}(Q)} - 1,$$

$$\mathrm{Basis}^{DW,\$}_{c,t}(Q) = P^{DW}_{c,t}(Q) - P^{DW}_{\mathrm{USD},t}(Q),$$

and report analogous measures based on mid- and micro-prices for comparison.

For each of the ten defined basis measures, we test whether the basis has non-zero mean by applying the Newey–West heteroskedasticity and autocorrelation consistent (HAC) t-test to the log-transformed percentage basis. For each basis, we reject the null hypothesis of zero mean at the $1\%, 5\%, 10\%$ significance levels, concluding that BTC/USD were traded at a price statistically different from BTC quoted in the listed stablecoins. Generalized Pareto Distribution framework Park and Kim, 2016 is applied to estimate and analyze the tail index ξ to access the extremeness of the deviations. For all basis we obtained the result $\xi < 0$, which indicated bounded tails, implying that the deviations do not exhibit explosive or infinite-variance behavior. We note that the estimated ξ for USDC specifications are closer to 0 than those of USDT, indicating that the USDC market exhibits a significantly higher probability of large dislocations during stress event.

We test stationarity using the KPSS test (H_0: basis is stationary) and nonstationarity using the ADF test (H_0: basis has unit root). We consistently reject null for KPSS test, however we do not have sufficient evidence to conclude all basis contain unit roots. This scenario often indicates the existence of structural breaks in data structure and reveals that the basis dynamics are regime-dependent rather than universal (Maddala and Kim, 1998). Given this result, we apply the Binary Segmentation method to the USDT basis to detect two critical change points for level shift: $t_1 = 2023\text{-}03\text{-}10\ 22{:}11{:}00$, $t_2 = 2023\text{-}03\text{-}13\ 03{:}31{:}00$. These breakpoints partition the sample into three regimes: pre-stress, stress, and post-stress. All subsequent analysis is performed both on the full sample and separately within each regime.

We evaluate whether the statistically significant basis deviations exceed total transaction costs at traded size Q. We define the size-dependent total execution cost for an arbitrage direction that buys i and sells in j, where $i, j \in \{USD, c\}, c = USDT, USDC$ as:

$$\begin{aligned}\tau^{i\to j}_t(Q) = &\left(P^{DW,\mathrm{buy}}_{i,t}(Q) - P^{\mathrm{mid}}_{i,t}\right)\\ &+ \left(P^{\mathrm{mid}}_{j,t} - P^{DW,\mathrm{sell}}_{j,t}(Q)\right)\\ &+ f_i P^{DW,\mathrm{buy}}_{i,t}(Q) + f_j P^{DW,\mathrm{sell}}_{j,t}(Q), \quad i \neq j.\end{aligned}$$

Here f_c is a proportional taker fee, for Kraken, this is assumed to be $f_c = 0.0004$ to approximate high-volume institutional execution. We also show results with $f_c = 0.001$ on the full sample to show effects of taker fees.

We define the tradability indicator $I_t(Q) = \mathbf{1}\left\{|\mathrm{Basis}^{DW,\$}_t(Q)| > \tau_t(Q)\right\}$ and report the tradable frequencies of deviations conditioned on the time when Q BTC are tradable and unconditioned on the total time period. It is shown that for both USDC and USDT markets, deviations were not arbitrages before the SVB event. For USDT and a small trading size, basis deviations overall exceed no arbitrage bounds for more than 50% of the time and frequency decreases as the traded size increases. These arbitrages accumulated during and after the SVB event. Another noticeable observation is that for USDC and a large traded size, conditioned tradable frequency reaches 75%. However these arbitrage opportunities only appeared during 2% of the time in total trading periods, which took place during the SVB event, reflecting that sufficient depth for large trades is rarely available during stress. In general, the USDC market exhibits less arbitrage opportunities across all traded sizes and regimes.

Table 1: Tradable Frequency Under Different Taker Fee

		$f = 0.0004$		$f = 0.001$	
ALT	Q	Cond.	Uncond.	Cond.	Uncond.
USDC	1	0.163	0.152	0.128	0.120
USDC	5	0.176	0.088	0.164	0.082
USDC	10	0.752	0.026	0.737	0.025
USDT	1	0.526	0.504	0.436	0.418
USDT	5	0.498	0.398	0.342	0.273
USDT	10	0.367	0.084	0.249	0.057

Table 2: Tradability Frequency by Regime ($f = 0.0004$)

ALT	Q	Pre-SVB	During SVB	Post-SVB
USDC	1	0.003 / 0.001	0.993 / 0.099	0.140 / 0.053
USDC	5	0.002 / 0.001	0.994 / 0.072	0.256 / 0.015
USDC	10	0.008 / 0.000	0.997 / 0.024	0.535 / 0.002
USDT	1	0.004 / 0.002	0.997 / 0.111	0.989 / 0.391
USDT	5	0.002 / 0.001	0.974 / 0.088	0.959 / 0.310
USDT	10	0.001 / 0.000	0.951 / 0.016	0.915 / 0.068

2.1 Basis Persistence Analysis After Transaction Costs

In this section we examine persistence of cross-currency basis deviations after accounting transaction cost. Let $\tau_t^+(Q)$ denote the boundary for the direction of buying in USD and selling in $c \in \{USDT, USDC\}$ and $\tau_t^-(Q)$ the opposite direction. We define the transaction cost adjusted residual

$$g_t(Q) = \begin{cases} \text{Basis}_t^{DW}(Q) - \tau_t^{(+)}(Q), & \text{Basis}_t^{DW}(Q) \geq 0, \\ \text{Basis}_t^{DW}(Q) + \tau_t^{(-)}(Q), & \text{Basis}_t^{DW}(Q) < 0. \end{cases}$$

$g_t(Q)$ measures economically exploitable deviations net of execution costs.

We first report the distributions of $g_t(Q)$. For USDT, the overall means of the residuals for all traded size are negative, thus on average the deviations remain within the boundary, although $P(g > 0)$ confirms that arbitrage opportunities occur. For USDC, especially during the SVB event, both the mean residuals and $P(g > 0)$ are larger for all traded sizes, reflecting severe and episodic dislocations during the stress period. Notably, during the stress period $P(g > 0)$ approaches unity for USDC, implying nearly deterministic arbitrage.

Table 3: Means of Cost-Adjusted Residuals

Q	Fee	ALT	Mean	$P(g > 0)$
1	0.0004	USDT	-34.85	0.290
	0.0004	USDC	86.94	0.452
	0.0010	USDT	-16.33	0.379
	0.0010	USDC	74.87	0.418
5	0.0004	USDT	-27.39	0.315
	0.0004	USDC	110.95	0.445
	0.0010	USDT	-8.92	0.471
	0.0010	USDC	98.57	0.433
10	0.0004	USDT	-18.78	0.356
	0.0004	USDC	542.80	0.849
	0.0010	USDT	-5.80	0.473
	0.0010	USDC	522.02	0.833

Table 4: Mean Cost-Adjusted Residuals During SVB Stress ($f = 0.0004$)

Q	ALT	Mean	$P(g > 0)$
1	USDT	-144.65	0.003
	USDC	845.26	0.993
5	USDT	-130.03	0.026
	USDC	805.77	0.994
10	USDT	-114.35	0.049
	USDC	765.16	0.997

We perform survival analysis on the duration of an excursion $I_t = 1\{g_t > 0\}$ outside the no-arbitrage band, treating it as a random variable T, and estimate the survival function $S(t) = \mathbb{P}(T > t)$ using the Kaplan–Meier estimator. We also compute the half-life of residuals using an AR(1) estimate, where $\hat{\phi}(Q)$ denotes the autoregressive coefficient at trade size Q, to measure the time required for a deviation to decay by half. Across both USDT and USDC markets, the median duration of tradable excursions is approximately two seconds, and half-lives are short prior to the stress event, indicating strong mean reversion and rapid correction of arbitrage opportunities. During the stress event, however, the two markets diverge sharply: mean excursion durations for USDC spike to more than 30 minutes and the half-life approaches near unit-root behavior, with autoregressive coefficients exceeding 0.999 and half-lives surpassing two hours at $Q = 5$. In contrast, although USDT deviations also become more persistent during stress, their magnitude remain substantially smaller than those of USDC. After the SVB event, both duration and half-life measures for USDC decline but remain longer relative to pre-SVB levels, while USDT deviations revert more quickly and, in terms of duration, even fall below their pre-stress levels, confirming faster correction dynamics in the USDT market.

Table 5: Excursion Duration Statistics: Fee Comparison

		Fee = 0.0004		Fee = 0.0010	
ALT	Q	N	Mean (s)	N	Mean (s)
USDC	1	79756	8.996	76047	8.735
USDC	5	37543	10.075	36638	10.043
USDC	10	866	56.364	928	51.653
USDT	1	27754	16.981	48177	12.795
USDT	5	40724	10.480	62305	10.242
USDT	10	13775	10.025	14863	12.373

Table 6: Excursion Count / Mean Duration by Regime

ALT	Q	Pre (N/Mean)	During (N/Mean)	Post (N/Mean)
USDC	1	51613/6.78	72/2306.25	28073/7.17
USDC	5	34964/6.38	59/2072.25	2522/13.09
USDC	10	760/7.34	36/1126.72	71/37.58
USDT	1	24500/18.92	174/3.60	3080/2.34
USDT	5	32591/12.29	477/8.18	7656/2.93
USDT	10	10666/11.81	139/10.23	2970/3.60

Table 7: AR(1) Persistence and Half-Life (Full Sample)

ALT	Q	$\hat{\phi}$	Half-Life (s)
USDT	1	0.9889	62.37
USDT	5	0.9737	25.98
USDT	10	0.9501	13.53
USDC	1	0.9984	444.44
USDC	5	0.9983	402.44
USDC	10	0.9997	2039.75

Table 8: AR(1) Persistence by Regime

		Pre		During		Post	
ALT	Q	ϕ	HL	ϕ	HL	ϕ	HL
USDC	1	0.7133	2.05	0.999917	8392.99	0.9033	6.82
USDC	5	0.7496	2.40	0.999928	9594.52	0.9442	12.07
USDC	10	0.6854	1.84	0.999896	6680.89	0.9697	22.54
USDT	1	0.8428	4.05	0.999233	903.69	0.9903	71.47
USDT	5	0.7701	2.65	0.998699	532.28	0.9834	41.43
USDT	10	0.7859	2.88	0.998221	389.26	0.9816	37.40

All analysis consistently points to the existence of statistically significant deviations between BTC prices quoted in USD and stablecoins, and these deviations are multi-aspect, regime- and trade-size dependent. During the SVB event, USDC experienced a substantial breakdown in arbitrage efficiency. While arbitrage opportunities were extremely rare, once tradable deviations appeared, they persisted for an exceptionally long time. In contrast, USDT markets exhibited substantially faster correction dynamics. Arbitrage opportunities occurred more frequently, but deviations decayed quickly during and after the SVB event. Note that these observations hold true when we increase the taker fees to 0.001.

In addition, we perform the same set of analysis using Coinbase USD/USDT pair and Binance USDC/USDT pair. The taker fee for Coinbase is assumed to be 0.0004 and for Binance is 0.00001. Our first observations are that both Coinbase and Binance offer lower liquidity than Kraken, so $Q = 10$ trades were mostly unavailable. For Binance the time period with level shift is 2023-03-11 01:32 to 2023-03-12 22:47 indicating that the SVB event had a shorter impact on Binance's BTC prices, and the time period for Coinbase is identical to Kraken. The statistical results are similar to those of Kraken data with the USD/USDT pair. The only notable differences are that BinanceUS exhibited 3% arbitrage opportunities for small trade size across the whole sample even when its taker fee is significantly lower. Both exchanges give the same results from the persistence analysis from the perspective of arbitrage correction speed. We conclude that the observed deviations dynamic was universal during March 2023.

3 Liquidity, Fragmentation, and Volatility

3.1 Section Methodology

Following Brauneis et al., 2022, to measure liquidity, depth, and spread, we implement the following metrics:

$$\text{EffectiveSpread}_t^{(\text{bps})} = 10^4 \cdot \frac{2|p_{i,t}^{\text{trade}} - \text{mid}_t|}{\text{mid}_t} \quad (1)$$

$$\text{QS}_{i,t} = 10^4 \cdot \frac{P_{i,t}^{\text{ask},1} - P_{i,t}^{\text{bid},1}}{\text{mid}_t} \quad [\text{bps}] \quad (2)$$

$$\text{BSlip}(Q) = \frac{\text{VWAP}^{\text{buy}}(Q) - \text{mid}_t}{\text{mid}_t} \quad (3)$$

$$\text{SSlip}(Q) = \frac{\text{mid}_t - \text{VWAP}^{\text{sell}}(Q)}{\text{mid}_t} \quad (4)$$

$$\text{RTC}(Q) = 10^4 \cdot \text{BSlip}(Q) + 10^4 \cdot \text{SSlip}(Q) \quad (5)$$

$$\text{Depth}_{10\text{bps}} = \sum_{p_{i,t} \in [\text{mid}_t \cdot 0.999, \text{mid}_t \cdot 1.001]} q_{i,t} \quad (6)$$

Where $\text{VWAP}^{\text{buy}}(Q)$ is the volume-weighted average price of sweeping Q USD of notional through the ask side, and $\text{VWAP}^{\text{sell}}(Q)$ is from sweeping the bid side. $\text{RTC}(Q)$ represents Round-Trip Cost, $q_{i,t}$ represents BTC amount at level i, $p_{i,t}$ represents the bid or ask at level i, and mid_t represents the mid quote at t.

For each pair (h, m) of bucket hour and market id, the hourly realized variances $\text{RV}_{m,h}^{\text{disp}}$ $\text{RV}_{m,h}^{\text{USD}}$ are simply the sum of squared minute intraday log returns. We construct minute intraday log returns and USD-normalized log returns from mid quotes at the market level; return series are reset at midnight UTC to avoid overnight volatility contamination. Peg returns are defined as the minute-by-minute change in each stablecoin's USD value at a given timestamp. For each bucket hour h and quote q, markets are aggregated using an equal-weighted mean across all markets for a given quote denomination. USD-common realized variance for USDT and USDC stablecoins decomposes as (Ahmed et al., 2024):

$$\mathrm{RV}^{\mathrm{USD}}_{m,h} = \underbrace{\sum_{t\in h} r^2_{m,t}}_{\mathrm{RV}^{\mathrm{BTC/s}}_{m,h}} + \underbrace{\sum_{t\in h} \left(r^{\mathrm{peg}}_{m,t}\right)^2}_{\mathrm{RV}^{\mathrm{peg}}_{m,h}} + \underbrace{2\sum_{t\in h} r_{m,t}\cdot r^{\mathrm{peg}}_{m,t}}_{\mathrm{COV}_{m,h}} \quad (7)$$

For this section, we define regimes as non-overlapping fixed date periods, where the stress regime incorporates the USDC depeg event: Pre (2023-03-01 → 2023-03-09, 216 hours), Stress (2023-03-10 → 2023-03-13, 96 hours), and Post (2023-03-14 → 2023-03-21, 192 hours). Since the depeg stress event is a dated and known public information shock, an event-study style regime classification provides a sufficient sandbox for the quote currency liquidity and volatility analysis in the following sections.

3.2 Volatility

Looking only at BTC-based realized variance (RV) depicted in Figure 1, we see that during the stress regime, USDC RV spiked in alignment with its depeg following the SVB collapse. However, as the peg is restored, BTC-based realized volatility consolidates to approximately the same levels for all quote currencies. This conclusion is further reinforced by MBB[1] with block size $L = 12$ hours on BTC-based volatility gaps for common exchanges. We find that while volatility gaps exist throughout each regime, only the gaps during the stress regime for USDC are significantly different than USD volatility. These findings imply that while BTC-realized volatility gaps may exist due to exchange effects and high-frequency variance, these gaps are essentially inconsequential at lower timeframes (1hr, etc.). However, BTC-based volatility doesn't tell the full story. When we look deeper and attribute variance to specific components, we see that USDT and USDC have additional "peg variance", which as seen in Figure 2 can be extremely large during times of stress/depeg. The negative covariance for USDC during the stress regime also suggests that during the depeg, BTC/USDC returns and peg returns moved against each other. The peg variance begins to dematerialize after the stress event, yet Figure 2 still depicts an important fact; volatility is systematically different for BTC quoted in stablecoins versus BTC quoted in USD due to additional peg variance.

[1]MBB refers to Moving Block Bootstrap (Kunsch, 1989), which is a method for evaluating the significance of statistical estimates.

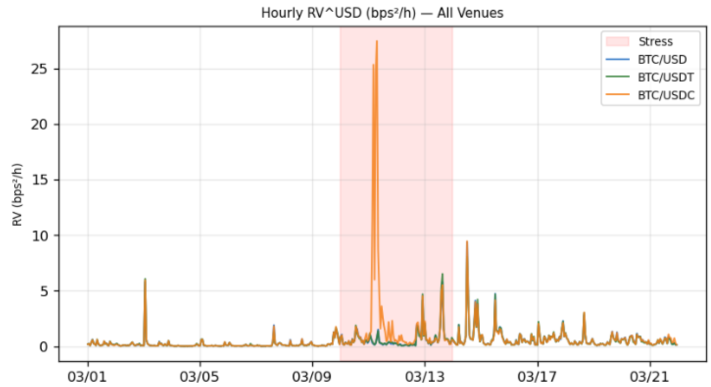

Figure 1: Hourly realized variance by quote and regime.

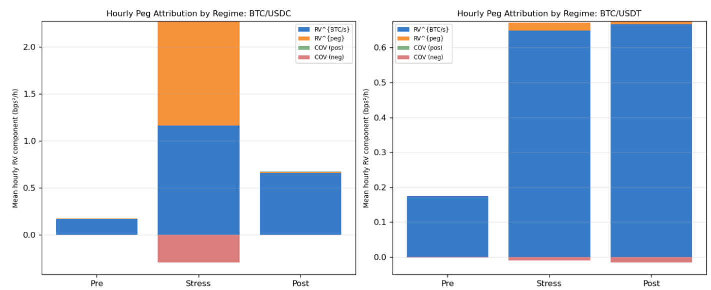

Figure 2: Hourly peg attribution: decomposition of USD variance into BTC/s, peg, and covariance components.

3.3 Liquidity

Figures 3, 4, 6, and 5 plot the aforementioned liquidity metrics for each quote currency across the three regimes. Across all figures, a consistent pattern emerges: stablecoin liquidity, depth, and execution cost metrics differ substantially from USD across all regimes. In the post-stress regime, spreads tend to tighten and depth increases for USDT and USD, while the opposite occurs for USDC—spreads widen and depth declines—suggesting potential liquidity shifts away from USDC following the depeg stress event into the other quote currencies. Round-trip cost and Slippage patterns further reinforce this interpretation: USDC becomes sharply more expensive during stress and remains elevated post-stress. The slippage curves for USDT and USD remain largely similar across trade sizes, suggesting relatively stable execution quality for those pairs. To ensure that our conclusions are valid and not just noise, we again use MBB with block size $L = 1$ hour and indeed verify that each metric per stablecoin, per regime, is different than the USD quoted counterpart with statistical significance[2]. Across pre and post regimes, BTC/USD has systematically tighter spreads, higher depth, and lower RTC than BTC quoted in stablecoins, implying systematic differences in liquidity versus stablecoins.

3.3.1 Effective Spread and Fragmentation

Effective spread (Figure 7) is interesting as it tells a somewhat different story than the quoted spread. Ef-

[2]We also calculated all metrics for only common exchanges containing data for all pairs as a robustness check and saw the same patterns for each metric.

fective spread increases during and post-stress, but even though USDT post-stress effective spread widens more than USD, the differences are fairly consistent across the two. USDC effective spread remains elevated post-stress, indicating that, while USDT and USD spreads equalize post-stress, traders still pay a premium to trade USDC post-stress. This fact shows that fragmentation across quote currencies may be more visible in quoted liquidity. Fragmentation can also be represented by crosses between bid and ask prices[3]. To identify fragmentation in each quoted currency, we compare quoted currencies to their respective consolidated order books across venues. Table 9 depicts the cross-frequency and mean magnitude of cross for each quote per regime. The table shows that cross-venue crossings are common in all quotes. During the stress regime, USD shows the highest crossing frequency with large crossing magnitude[4]. USDC crossing magnitude is enormous during stress, depicting the dislocation severity during the depeg. Combined with the previous conclusions that USD spreads are tighter and order books are deeper relative to USDT and USDC, these results may indicate a shift toward USD quote currency.

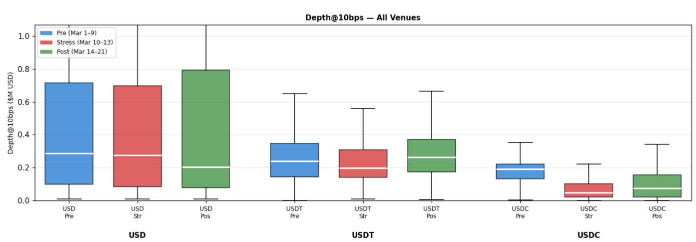

Figure 3: Depth at 10 bps across regimes and quotes.

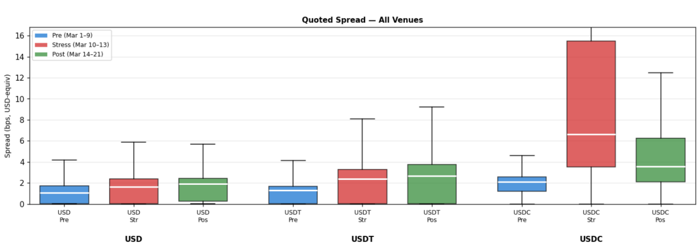

Figure 4: Quoted bid–ask spreads by quote and regime.

4 Drivers of Cross-Currency Basis Deviations

For the current analysis, we construct a high-frequency microstructure-driven multiple linear regression model to explain short-horizon movements in all pairwise basis computations for Bitcoin, quoted in USD, USDT, and USDC. Starting from the Kraken exchange level L2 snapshots, we collapse the order book to second-by-second top of book and depth within band measures (e.g., best bid/ask prices, spreads, total depth, and bid–ask

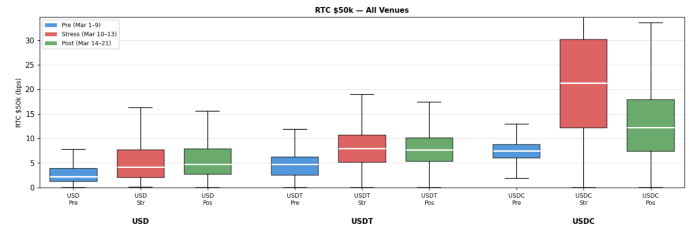

Figure 5: Round-trip cost for $50k trade size by quote and regime.

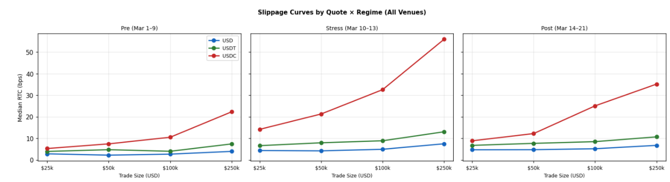

Figure 6: Slippage curve for increasing trade sizes.

imbalances), both at the touch and within narrow price bands such as 1–5 bps around the midprice, to capture liquidity that is either immediately actionable or deeper in the book. In our variable selection process for the model, we decided to implement the criterion revolving the partial f-statistics of the regressors. Particularly, allowing x_j to represent the most recent regressor added to the model, we enact the F-test of $H_0 : \beta_j = 0$; which is regarded as the partial F-statistic associated with x_j. Because many high-frequency market microstructure variables are mechanically and economically correlated, regressors were first grouped into four bins representing distinct market mechanisms to limit multicollinearity, with only one variable selected from each bin. The chosen regressors are the spread differential between markets, 60-second rolling basis volatility, the 5 basis point depth band imbalance differential between markets, and a lead–lag term. The dependent variable, the first difference of the log cross-currency basis, focuses the analysis on short-horizon, transitory adjustments rather than long-run level differences. In summation, we estimate a sequence of lagged linear regressions for all pairwise basis comparisons for lag lengths $\ell \in 1, 3, 5, 10, 15, 30, 60, 120, 300$, in seconds. Our results from this analysis are represented in the table below. All results are statistically significant at the 95% level across all lags.

Across all currency pairs, the spread differential has a statistically significant but economically small effect, with its direction varying by pair, negative for USDT/USD and USDC/USDT, and positive for USDC/USD. Basis volatility is the dominant regressor in all specifications, though its sign differs: strongly negative for USDT/USD (around -0.63) and positive for USDC/USD and USDC/USDT (around 0.13–0.16). The magnitude of the volatility effect is substantially larger for USDT/USD than for the other pairs, indicating a much stronger sensitivity of that basis to past volatility.

[3] Following Jeon et al., 2021 cross-venue consolidated order book crossings (frequency/magnitude) quantify dislocations consistent with fragmented liquidity across venues.

[4] Consistent with Makarov and Schoar, 2020, stress increases frictions and slows arbitrage linking venues.

Table 9: Cross-venue crossing frequency and magnitude (Table A, 60s)

Quote	Regime	Book	N snapshots	Cross freq.	Cross mag. mean
USD	pre	consolidated_USD	12,955	0.5135	2.1665
USD	stress	consolidated_USD	7,200	0.7835	14.3267
USD	post	consolidated_USD	10,057	0.5960	7.4787
USDC	pre	consolidated_USDC	12,835	0.2108	1.0080
USDC	stress	consolidated_USDC	7,174	0.5827	47.6233
USDC	post	consolidated_USDC	10,019	0.3857	2.6697
USDT	pre	consolidated_USDT	12,958	0.7030	2.3832
USDT	stress	consolidated_USDT	7,200	0.7067	4.6747
USDT	post	consolidated_USDT	10,080	0.6415	4.4932

	OLS Analysis, Dependent Variable: $\Delta \log(\text{Basis}_t)$											
	USDT/USD				**USDC/USD**				**USDC/USDT**			
Lag (s)	SpreadDiff	Vol	ImbDiff	Lead-Lag	SpreadDiff	Vol	ImbDiff	Lead-Lag	SpreadDiff	Vol	ImbDiff	Lead-Lag
1	-7.97×10^{-5}	-0.6285	5.0×10^{-4}	-5.0×10^{-4}	6.22×10^{-5}	0.1306	3.0×10^{-4}	0.0153	-2.58×10^{-5}	0.1609	0.0023	0.0143
3	-7.93×10^{-5}	-0.6379	5.0×10^{-4}	-5.0×10^{-4}	5.98×10^{-5}	0.1339	3.0×10^{-4}	0.0153	-2.70×10^{-5}	0.1650	0.0024	0.0144
5	-8.00×10^{-5}	-0.6367	5.0×10^{-4}	-5.0×10^{-4}	5.97×10^{-5}	0.1331	3.0×10^{-4}	0.0154	-2.70×10^{-5}	0.1644	0.0024	0.0144
10	-7.93×10^{-5}	-0.6278	5.0×10^{-4}	-5.0×10^{-4}	6.05×10^{-5}	0.1322	3.0×10^{-4}	0.0153	-2.62×10^{-5}	0.1624	0.0024	0.0144
15	-7.98×10^{-5}	-0.6257	6.0×10^{-4}	-6.0×10^{-4}	6.04×10^{-5}	0.1330	3.0×10^{-4}	0.0154	-2.87×10^{-5}	0.1628	0.0024	0.0144
30	-7.94×10^{-5}	-0.6218	6.0×10^{-4}	-5.0×10^{-4}	6.16×10^{-5}	0.1287	4.0×10^{-4}	0.0153	-2.57×10^{-5}	0.1590	0.0025	0.0144
60	-7.90×10^{-5}	-0.6234	6.0×10^{-4}	-5.0×10^{-4}	6.06×10^{-5}	0.1333	4.0×10^{-4}	0.0153	-2.59×10^{-5}	0.1640	0.0025	0.0144
120	-7.88×10^{-5}	-0.6259	7.0×10^{-4}	-5.0×10^{-4}	6.09×10^{-5}	0.1297	4.0×10^{-4}	0.0154	-2.52×10^{-5}	0.1600	0.0025	0.0145
300	-7.90×10^{-5}	-0.6267	7.0×10^{-4}	-5.0×10^{-4}	6.19×10^{-5}	0.1342	5.0×10^{-4}	0.0151	-2.38×10^{-5}	0.1646	0.0025	0.0142

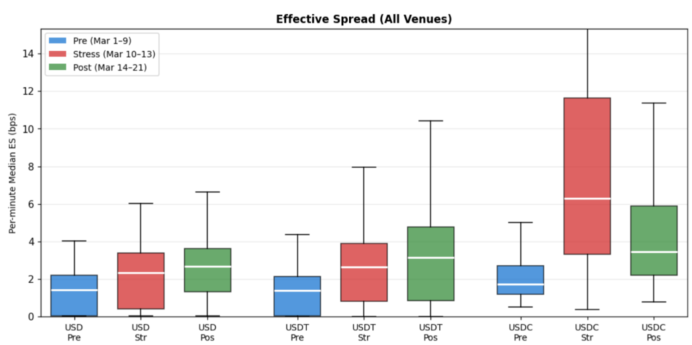

Figure 7: Effective spread by quote and regime.

Order book imbalance is consistently positive across all pairs and increases modestly as the lag lengthens, suggesting that directional pressure accumulates over time. Finally, lead–lag effects are pair-specific but stable across lags, negative for USDT/USD and positive for both USDC-based pairs, with larger magnitudes in the USDC/USD and USDC/USDT relationships. In addition to the previously listed results, we performed an analysis of residuals for each model at all lags and received residuals with a first moment of zero and a second central moment of one after standardization, validating the assumptions of the OLS regression models. Lastly, using the hat matrix, we computed Cook's distance for each observation across all models and lags and discovered that approximately 57% of the influential observations used to estimate the parameter vector β came during the period 03/11/2023 - 03/13/2023, corresponding to the period of maximal stress in our evaluated snapshot.

5 Regime Effects and their Predictors

In the current analysis, we model stress regime transitions using a generalized linear model framework. More specifically, we construct and test a logistic regression model to evaluate the best predictors of regime change in a given cross-currency pairing. Our variable selection process follows the same structure as in the former OLS analysis; however, we ensure not to confound our chosen regressors with our stress regime features. We define regimes using K-means clustering, an unsupervised learning method used to partition observations into K disjoint groups based on similarity in a chosen feature space. Each cross-currency pair (e.g., BTC/USD vs. BTC/USDT, or more generally, quote A vs. quote B) is treated as its own microstructure system. For a given pair, regimes are defined using cross-market basis volatility together with contemporaneous transaction cost and deep in the book depth measures from the leading stablecoin quoted market. This specification is motivated by the fact that stablecoin markets serve as the primary liquidity and execution venue for crypto assets, concentrating order flow and arbitrage activity even though the stablecoins themselves are pegged. Consequently, spreads and depth in the stablecoin market summarize prevailing execution frictions and effective arbitrage capacity, while basis volatility captures instability in the cross-currency linkage. This approach defines market stress based on whether liquidity is tight and trading is costly in the leading stablecoin market, rather than by averaging conditions across both markets in a way that could obscure the true source of instability. The regressors are constructed from lagged microstructure measures that capture asymmetric order flow pressure across quote currencies, differentials in relative execution costs across venues, and delayed cross-market price adjustments. The lag length is set to five seconds ($\ell = 5$), so the model evaluates whether current microstructure conditions contain predictive information about the probability of entering a stressed liquidity regime five seconds into the future. This structure preserves a clean separation between contemporaneous regime definition and forward-looking regime prediction. All regressors are standardized to have zero mean and unit variance prior to estimation, so coefficient magnitudes are directly comparable across variables and reflect the effect of a one standard deviation change in each microstructure predictor on the probability of a regime transition. Our results are as follows:

Currency Pair	Metric	Value
USDT/USD	Spread (Stress/Calm)	8.10 / 2.41
	Depth (Stress/Calm)	4.96 / 12.74
	ImbalanceDiff β / P-value	-0.0453 / 0.003
	SpreadDiff β / P-value	0.0126 / 0.007
	Lead-Lag β / P-value	-1.2165 / 0.142
	ROC-AUC	0.553
USDC/USD	Spread (Stress/Calm)	11.51 / 0.42
	Depth (Stress/Calm)	2.25 / 99.41
	ImbalanceDiff β / P-value	1.0214 / 0.000
	SpreadDiff β / P-value	-0.0319 / 0.000
	Lead-Lag β / P-value	5.7344 / 0.000
	ROC-AUC	0.820
USDC/USDT	Spread (Stress/Calm)	11.51 / 0.42
	Depth (Stress/Calm)	2.25 / 99.41
	ImbalanceDiff β / P-value	2.5796 / 0.000
	SpreadDiff β / P-value	-0.0347 / 0.000
	Lead-Lag β / P-value	8.0456 / 0.000
	ROC-AUC	0.948

The logistic regression results, estimated with cluster-robust standard errors, evaluate predictors of stress transitions in the USDT/USD, USDC/USD, and USDC/USDT pairings during the 02/28/23–03/23/23 stablecoin crisis. Liquidity metrics show that USDT markets remained relatively stable, with modestly higher spreads and slightly lower depth during stress, whereas USDC experienced extreme liquidity shocks: very wide spreads and shallow depth in stress periods compared to calm periods. Predictor effects vary across markets: in the USDT/USD pairing, imbalance differentials between markets have a small negative effect on stress probability, while in USDC pairings, it is strongly positive, reflecting the role of imbalances in signaling stress during the USDC depeg. Changes in spread differentials affect markets differently too: when spreads widen in USDC relative to other markets, the resulting differential is a reliable indicator of stress, while spread differentials in USDT/USD tend to capture short-lived execution frictions across venues rather than sustained liquidity impairment. Lead-Lag predictors are insignificant for USDT/USD but strongly positive in USDC/USD and USDC/USDT, capturing the rapid propagation of shocks. Model performance mirrors these patterns: ROC-AUC is low for USDT/USD (0.553) but high for USDC/USD (0.820) and USDC/USDT (0.948), indicating that predictors are most informative in markets directly impacted by the depeg. Overall, these results highlight that market microstructure variables such as imbalance and spread differentials, and lead-lag effects provide significant predictive power for regime change during periods of extreme stablecoin stress, particularly in markets directly influenced by the event.

5.1 Testing for Differences Across Regimes

We now evaluate how high-frequency microstructure variables impact the evolution of the log basis for different currency pairs under different stress regimes. Specifically, utilizing the K-means classification technique, we estimate regime-dependent lagged effects of key explanatory variables using a pooled ordinary least squares regression with heteroskedasticity and autocorrelation consistent (HAC) standard errors. The set of regressors employed in this analysis is identical to that used in the baseline OLS specification for changes in the log basis, ensuring that differences in estimated coefficients are attributable solely to regime dependence rather than changes in model inputs. Each explanatory variable is lagged over multiple horizons to capture both short and long-term dynamics, and each lagged regressor is interacted with a stress indicator, allowing its marginal effect to differ between calm and stress periods. Statistical inference is conducted using individual Wald tests to assess whether specific predictors exhibit statistically significant regime shifts, along with a joint Wald test evaluating whether the full set of lagged microstructure effects differs systematically across regimes.

Across all quote currencies, estimated betas exhibit clear regime dependence, with both signs and magnitudes shifting between calm and stress periods in lag-specific ways. For USDT/USD, calm period betas for Spread Diff and Basis Vol are consistently negative, while stress adjustments are positive at very short lags (1–5) and slightly negative at longer lags (30–300), partially offsetting or reinforcing calm effects. For USDC/USD, calm betas are generally small, but stress effects are pronounced and vary by lag: Spread Diff stress betas are mostly negative (with small positives at medium lags), Basis Vol stress betas are positive at all lags with greater magnitude than the calm betas, and Lead–Lag stress effects increase at most lags from the calm beta magnitudes. USDC/USDT shows similar dynamics to the USDC/USD betas across regimes and lags, although most calm period betas for Imbalance Diff in the USDC/USDT analysis are positive as opposed to negative. Overall, USDC/USD is most sensitive to stress shocks in the aggregate, USDT/USD maintains the strongest calm period predictability, and all variables are jointly significant in explaining regime-dependent log-basis dynamics.

6 Exchange Effects on the BTC-USDC/USDT Basis

In the current analysis, we examine the BTC–USDC/USDT basis across the Kraken, BinanceUS, and Bybit exchanges using our previously defined OLS regression framework. Each exchange is modeled independently, ensuring that all regressors and lags are computed solely from that exchange's order book. Coefficients $\hat{\beta}_i$ with Newey-West HAC standard errors are extracted, and pairwise comparisons are performed to identify variables whose effects differ significantly across venues. Our analysis relies on two key assumptions: (i) independence across exchanges, reflecting the distinct microstructure and noise of each order book, and (ii) HAC standard errors account for autocorrelation and heteroskedasticity within each exchange but not across exchanges. This section aims to illustrate which microstructure effects are similar across exchanges and which vary depending on the exchange.

Across most variables and most lags considered, a clear and highly stable ordering of coefficient magnitudes emerges:

$$\text{Kraken} > \text{BinanceUS} > \text{Bybit}$$

This ordering holds for most regressors, lags, and pairwise comparisons, with nearly all differences statistically significant at the 95% level. Across all lags, Kraken consistently exhibits the largest coefficients, BinanceUS is intermediate, and Bybit is near zero or occasionally negative. Most pairwise differences between exchanges are highly significant, with many p-values effectively zero. Overall, the consistent dominance of Kraken coefficients across all variables and lags indicates that our chosen regressors are most effective on Kraken, intermediate on BinanceUS, and weakest on Bybit in our current snapshot. The stability of these results across lags suggests structural, exchange-specific microstructure effects rather than transient noise. These findings demonstrate that observable market signals propagate into basis changes with exchange-specific elasticities, underscoring that exchange choice materially affects how microstructure dynamics transmit into prices.

7 Stablecoin Dynamics Across Exchanges and Regimes

Stablecoins such as USDC and USDT are designed to maintain a fixed peg to the U.S. dollar, typically at a 1:1 ratio. Because of this, any deviations from the peg should be short-lived. Let P_t denote the observed market price of a stablecoin, and define the deviation from the peg as

$$x_t = P_t - 1. \tag{8}$$

Under normal conditions, arbitrage and redemption mechanisms push the price back toward the peg, so we expect x_t to revert toward zero. This naturally leads to modeling stablecoin prices using mean-reverting processes. A simple way to model this behavior in continuous time is with the Ornstein–Uhlenbeck process. In reality, prices are observed at discrete time intervals. When we discretize the OU process over a time step Δt, we obtain

$$X_{t+\Delta t} = \mu + \phi(X_t - \mu) + \varepsilon_t, \tag{9}$$

where ε_t is a noise term and $\phi = e^{-\kappa \Delta t}$. This gives the relationship $\kappa = -\frac{1}{\Delta t}\ln(\phi)$, which allows us to estimate continuous-time behavior using discrete data. We therefore fit an AR(1) model of the form

$$x_t = c + \phi x_{t-1} + \varepsilon_t. \tag{10}$$

After fitting this model to USDT and USDC hourly data for both exchanges, Binance and Kraken, we can observe Table 10, the clear differences in price dynamics before and after the stress event across the different exchanges. We can see USDC on Kraken has nan values before the stress event. This occurs because the fitted $\phi < 0$, which implies that deviations exhibit sign-alternating behavior rather than persistent monotone reversion. Because of this the model suggests that the dynamics are characterized by rapid, oscillating mean reversion. This suggests that deviations are short-lived and do not persist in one direction. Since this stress event took place, we are able to observe these changes in the price dynamics; thus, stronger regulation should lead to smaller deviations and faster reversion back to the peg.

Table 10: Estimated mean-reversion speed (κ) and half-life before and after stress

Exchange	Coin	Before		After	
		κ	Half-Life	κ	Half-Life
Binance	USDT	0.120	5.758	0.0218	31.737
Binance	USDC	2.576	0.269	0.0202	34.268
Kraken	USDT	0.810	0.855	0.474	1.462
Kraken	USDC	nan	nan	2.601	0.266

7.1 Monte Carlo Simulation with Volume-Driven Shocks

To capture stablecoin behavior during high trading activity, we model price dynamics as a combination of mean reversion and volume-driven shocks. The mean-reverting component reflects the tendency of prices to return to the peg, while the shock component captures short-term deviations that increase with trading volume. We begin with the AR(1) model introduced earlier to model deviations from the peg where $\epsilon_t^{OU} \sim \mathcal{N}(0, \sigma_{OU}^2)$ represents the baseline noise estimated from the pre-stress regime, so that the simulation reflects normal market conditions absent structural disruption. To model the impact of trading activity, we use returns calculated from the price series, and then relate the magnitude of returns to trading volume using a simple regression:

$$|r_t| = c_0 + c_1\sqrt{V_t} + e_t, \tag{11}$$

where V_t is the trading volume. This captures the empirical relationship that larger trading volume leads to larger price movements. From the regression, we obtain an estimate of the expected magnitude of returns. To convert this into a scale parameter for simulation, we apply a constant rescaling:

$$\sigma(V_t) = (c_0 + c_1\sqrt{V_t}) \cdot K, \tag{12}$$

where $K = \sqrt{\frac{\pi}{2}}$ serves as a normalization factor linking absolute deviations to standard deviation, since our interval is now $[0, \infty]$. After fitting our AR(1) model for the necessary parameters we can then simulate price paths over the event window using both components. Starting from $x_0 = 0$, we iterate:

$$x_t = \phi x_{t-1} + \epsilon_t^{OU}, \tag{13}$$
$$P_t = 1 + x_t, \tag{14}$$
$$P_t \leftarrow P_t(1 + \epsilon_t^{ret}), \tag{15}$$

where $\epsilon_t^{ret} \sim \mathcal{N}(0, \sigma^2(V_t))$. This structure separates the mean-reverting behavior from short-term shocks while allowing trading volume to directly influence volatility. In periods of elevated volume, the model generates larger deviations from the peg, consistent with observed market behavior during stress events. After simulating 10,000 paths over the stress window, we can observe Figure 8, which shows a subset of the Monte Carlo paths produced. While the simulation incorporates both mean reversion and volume-driven shocks, the paths remain tightly clustered around the peg. Even during periods of elevated trading volume, the model does not generate large downward deviations comparable to the observed de-pegging event.

This suggests that, under the estimated dynamics, increased trading volume alone is not sufficient to produce a sustained deviation from \$1. Table 11 provides a statistical comparison between realized and simulated outcomes. The realized minimum price of 0.8707 is significantly lower than even the most extreme simulated outcomes. In contrast, the simulated 1% and 5% quantiles remain extremely close to the peg, at approximately 0.99935 and 0.99941, respectively.

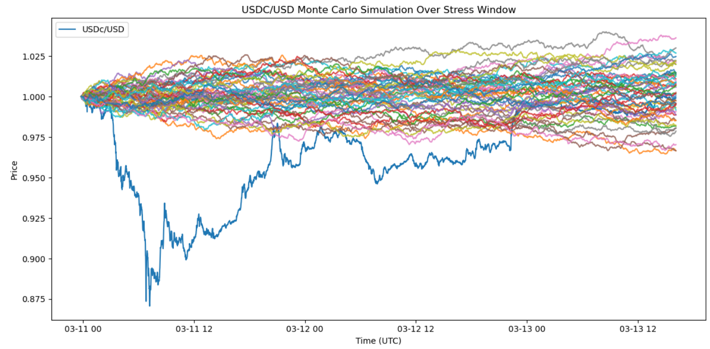

Figure 8: Monte Carlo simulation during the stress window.

Table 11: Realized vs. simulated distribution

Statistic	Value
Actual minimum price	0.87069543
Actual 5% quantile of minima	0.9100768806
Actual 1% quantile of minima	0.8885064732
Simulated 5% quantile of minima	0.9994108610
Simulated 1% quantile of minima	0.9993500152

The observed de-pegging event cannot be explained by volume-driven volatility alone and instead points to a breakdown in market structure, particularly in liquidity and confidence. This has direct implications for how regulation affects trading behavior.

Stronger reserve transparency and liquidity requirements, as proposed under the GENIUS Act, improve the reliability of redemption and reduce uncertainty about backing. This strengthens arbitrage activity across markets, leading to tighter alignment between stablecoin-quoted and USD-quoted prices and reducing persistent cross-currency deviations.

Within our framework, this would appear as faster mean reversion and fewer structural breaks. The Monte Carlo results show that, in the absence of such structural failures, even high trading volume does not generate large deviations from the peg. As a result, regulation improves not only stability but also overall market efficiency by reinforcing the mechanisms that keep prices anchored to \$1.

8 GENIUS Act on Stablecoins

During the SVB bank run, the absence of Cash Repos limited the direct availability of cash for primary arbitrage redemptions. The current utilization of Treasury Repos ensures an abundance of overnight cash liquidity, mitigating the risk of capital being locked in traditional bank accounts. Consequently, USDC's current asset composition closely aligns with the forthcoming GENIUS Act, which mandates strict 1:1 backing with high-quality liquid assets Circle Internet Financial, 2025.

In contrast, Tether (USDT) still holds approximately 23% of its reserves in non-compliant assets, including Bitcoin and precious metals Tether Holdings Limited, 2025. Tether's holdings do not currently meet GENIUS' mandated backing criteria which exposes them to being non-GENIUS classified. Recent projections by Citigroup suggest the stablecoin mar-

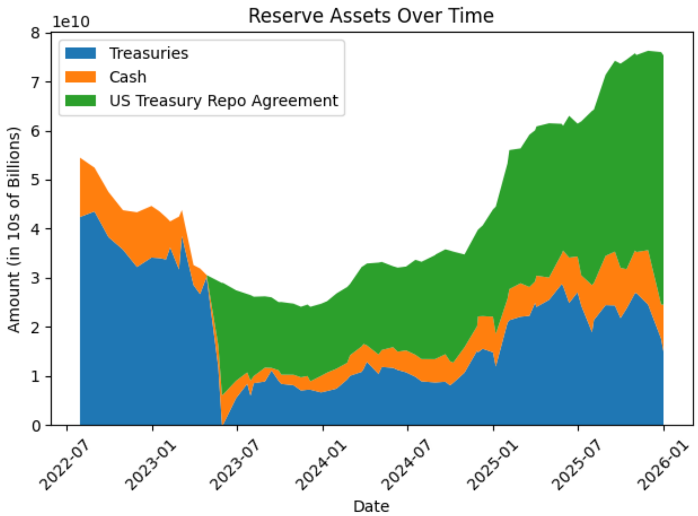

Figure 9: Circle's Reserve Asset Composition Pre- and Post-Shock.

ket will reach a base case of $1.9 trillion by 2030, with an optimistic bull case of $4.0 trillion Bantanidis et al., 2025. As stablecoins continue mature to trusted settlement layers their structural integrity is fundamental to traditional financial markets.

9 Conclusion

The former analysis illustrates that high-frequency market microstructure systematically drives short-horizon BTC cross-currency basis deviations across USD, USDT, and USDC. Basis volatility emerges as the dominant driver, strongly negative for USDT/USD and positive for USDC-based pairs, while spread differentials are economically small, imbalance effects accumulate modestly with lag, and lead–lag dynamics are stable but pair-specific. These microstructure elasticities vary across exchanges, with Kraken consistently exhibiting the largest coefficients, BinanceUS intermediate, and Bybit the weakest, reflecting persistent venue-specific differences in liquidity and order book structure. Effects are also regime-dependent: USDT/USD shows the most predictable behavior in calm periods, whereas USDC pairs are highly sensitive during stress, with volatility, spread, and imbalance effects amplifying at short to medium lags. Furthermore, stress transitions can be forecasted using lagged microstructure signals, particularly in USDC markets, where imbalance, spread, and lead–lag metrics strongly predict liquidity breakdowns. Systematic differences across quote currencies are not limited to temporary basis dislocations: volatility, liquidity, and fragmentation all differ between BTC quoted in USD versus stablecoins. When volatility is decomposed into BTC-specific and peg-related components, it becomes clear that even outside the stress regime, the additional peg variance makes stablecoin variance systematically different from USD-quoted variance. BTC quoted in USD also exhibited consistently tighter quoted spreads, greater depth, and lower round-trip costs across regimes than BTC quoted in USDT and USDC. While USD and USDT mostly recovered after the stress event, USDC continued to exhibit weaker liquidity and higher trading costs in the post-stress regime. These findings are reinforced by the fragmentation results: cross-venue dislocations were present in all quote currencies, but were most severe for USDC, while USD showed elevated crossing activity and large cross magnitudes during stress, with cross frequency remaining elevated post-stress. Consistent with the literature, the evidence suggests that quote currency choice materially affects measured volatility, liquidity, and fragmentation, especially during periods of market stress.

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